

PASIDU ILAMPERUMA

AI/ML Intern | Software Engineering Intern

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Pasidu.Mihiranga

Pasidu-Mihiranga

Portfolio

SUMMARY

Information Technology undergraduate focused on AI, Machine Learning, Full-Stack, and Cloud-Native Development. **Seeking internship opportunities in AI Engineering, Machine Learning Engineering, Applied AI Development.** Built and deployed projects involving large language models, computer vision, microservices, distributed systems, and modern web technologies. Additionally interested in **DevOps Engineering, Full-Stack Development, Game Development, Backend Systems, and Mobile Application Development.** Passionate about building scalable, intelligent, and impactful software solutions.

AWARDS ACHIEVEMENTS

- **Champion - AI Challenge Sri Lanka 2026**, Team Gmora, Innovative Ideathon and Buildathon, Organized by IEEE Young Professionals Sri Lanka.
- **Champion - BITCode 2026**, Team Gmora, Hackathon and Coding Contest, Department of Information Systems, Faculty of Management Studies, Rajarata University of Sri Lanka
- **Champion - InnovateX Hackathon 2025**, Team Gmora, Organized by Zebra Technologies
- **First Runner-Up - Vectra**, Team InsightAI King of the Hill CTF Competition, Organized by IIT Sri Lanka
- **Second Runner-Up - AlgoArena 2025**, Team Gmora, Leo Club of University of Sri Jayewardenepura
- **4th Place - ICS 3.0**, Team InsightAI, CTF Competition, Organized by University of Colombo School of Computing
- **4th Place - Genesiz,26**, Team Gmora, Ideathon and Buildathon, Organized by University of Colombo School of Computing
- **6th Place - JapuraXtreme 2025**, Team TrippleBoot, University of Sri Jayewardenepura
- **Finalist - GenZipher 2026, HaXtreme 2025, BioFusion 2025, CrptX 2026, Agentrix 2026, NeuroX 3.0**

PROJECTS

SuwaPath [Click to View](#)

2026

- Engineered a healthcare platform comprising a collection of specialized ML and vision models, orchestrated by a bounded LangGraph agentic loop and a 23-rule deterministic triage engine to achieve 100% emergency detection (8/8 cases).
- Developed a multilingual NLP pipeline mapping English, Sinhala, and Tamil symptom inputs to ~90 clinical concepts, enabling capability-aware matching to route patients to appropriate hospital specialties and facilities.
- Integrated predictive models for 7-day specialty demand forecasting and automated medical report OCR, benchmarking the multi-model ecosystem against a 114-vignette gold dataset and passing 150+ automated agentic scenario checks.
- **Stack:** Python, FastAPI, React, TypeScript, LangGraph, PostgreSQL, Qdrant, Tailwind

Purawasi [Click to View](#)

2026

- Built a voice-first multi-agent AI platform leveraging 20+ specialized agents to automate government service discovery, eligibility verification, document processing, workflow guidance, and citizen assistance in Sinhala, Tamil, and English.
- Designed a scalable LangGraph orchestration architecture with supervisor-based agent routing, hybrid RAG retrieval, OCR-powered document intelligence, cross-lingual memory, real-time web research, and community-driven knowledge enrichment while maintaining privacy through PII masking and RBAC.
- **Stack:** Python, FastAPI, LangGraph, Gemini 2.5 Flash, Llama 3.3 70B, Qdrant, PostgreSQL, React, React Native, TypeScript, Tesseract OCR, LiveKit, Whisper, Redis, Langfuse

FacelD Recognition [Click to View](#)

2026

- Developed a hybrid biometric authentication system utilizing ArcFace for face embedding extraction and a custom fine-tuned MobileNetV2 binary classifier for real-time presentation attack detection (liveness checking).
- Built a custom face processing pipeline using OpenCV CLAHE lighting normalization and PyTorch data augmentation, achieving an Equal Error Rate (EER) of 1.00% on the standard LFW benchmark.
- **Stack:** Python, PyTorch, OpenCV, ArcFace, MobileNetV2, NumPy, SQLite

Piyoway Tourist Guide (Live) [Click to View](#)

2026

- Built a tourism management platform that consolidates destination discovery, itinerary planning, reviews, and listings into a single system, replacing fragmented travel information sources through 4 user roles, administrative dashboards, cloud-based media management, and a PWA-enabled architecture.
- **Stack:** Next.js, TypeScript, NestJS, PostgreSQL, Prisma, Tailwind CSS, Docker, Nginx, Cloudinary

Auditra Click to View	2026
- Developed an Audit Valuation ERP platform with 10+ operational modules and 8 user roles, replacing fragmented spreadsheet-based audit workflows with a centralized system for compliance, risk assessment, and valuation management. Stack: Flutter, Django, PostgreSQL, REST API	
ShinnyMinds Click to View	2026
- Built a 3D exploration game featuring custom character movement, animation systems, and environment interactions, demonstrating real-world game development workflows using Unity and Blender through 4 major gameplay systems. Stack: Unity, C#, Blender	

EXPERIENCE

Co-Founder - Gmora	Feb 2026 – Present
<i>University Student Software Startup</i>	
Co-founded a student-led software startup, contributing to full-stack application development, software architecture, technical decision-making, project planning, and collaboration on hackathons and client-oriented software projects.	
Group Leader - Second Year Software Development Project	2026
<i>University of Moratuwa</i>	
Managed project planning, software development activities and team coordination for a large-scale software project.	
Group Leader - First Year Microcontroller Hardware Project	2025
<i>University of Moratuwa</i>	
Led a multidisciplinary team in designing and developing a microcontroller-based hardware project.	
Design Pillar Member	2024 – 2025
<i>Mathematics Society, University of Moratuwa</i>	
Active member of the Design Pillar and contributed to society activities and event branding.	

EDUCATION

University of Moratuwa	2023 – Present
<i>Bachelor of Science (Hons) in Information Technology</i>	
<i>CGPA - 3.71/4.0</i>	
Richmond College, Galle	2022
<i>G.C.E. Advanced Level - Physical Science Stream</i>	
<i>Physics: A – Chemistry: A – Combined Mathematics: B</i>	
Richmond College, Galle	2019
<i>G.C.E. Ordinary Level</i>	
<i>9 A Passes including Mathematics, Science and English</i>	

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++, JavaScript, PHP, C#

Web Development: HTML, CSS, React.js, Next.js, Node.js, Express.js, Spring Boot, MongoDB, SQL, PostgreSQL

Mobile App Development:Flutter, React Native, Kotlin

Game Development:Unity Game Engine

Design Software:Adobe Photoshop, Canva, Adobe Illustrator, Figma

Operating Systems:Windows 11, Ubuntu/Linux

Languages:Sinhala (Native), English (Fluent)

Leadership & Communication:Event Management, Public Speaking, Team Leadership, Demonstrating

CERTIFICATIONS

- CGMA Finance Leadership Program (FLP – CIMA) – AICPA & CIMA (2024 – Present)
- Python Programming Course – SITEC (2023)
- Java Programming Course – SITEC (2023)
- Web Development Course – SITEC (2023)
- Python for Beginners & Web Design for Beginners – University of Moratuwa (2023)
- Cambridge Key English Test (KET) – Cambridge Assessment English (2018)

REFERENCES

Mr. Noyel Medawatta Technical Lead Zebra Technologies Lanka Pvt. Ltd. noyel.medawatta@zebra.com	Ms. Nipuni Premadasa Lecturer Department of Information Technology University of Moratuwa nipunip@uom.lk
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